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Cooperative Control of Multirotor UAVs with Disturbance Rejection Capabilities for Load Transportation

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ABSTRACT Unmanned Aerial Vehicles (UAVs) are key tools in disaster response because they are capable of operating in difficult-to-access environments and providing real-time information. Their applications include search and rescue missions, surveillance, monitoring inaccessible areas, and delivering supplies to difficult-to-reach locations. Moreover, tasks involving the transport of heavy or bulky loads are beyond the normal capabilities of a single UAV. Collaborative transportation using multiple drones has been proposed as an effective and practical solution. This work focuses on the joint transport of a load supported by cables by a group of UAVs, which requires precise waypoint ranges and force coordination to maintain stability. Here, a non-linear collaborative control strategy is proposed involving three essential components: a real-time estimator for cable forces and directions, speed controllers for each drone, and a formation/coordination controller for the entire group of UAVs. The purpose is to allow the entire system to reject load-generated disturbances and share the load-carrying effort. The closed-loop convergence of this system is theoretically demonstrated. Furthermore, the paper presents the experimental platform developed during this work to monitor, control, and performance-analyze a set of UAVs, allowing us to validate our proposed strategy in real experiments.

INDEX TERMS UAV, Load transport, Disturbance rejection, Collaborative control, Multi-robot system.

I. INTRODUCTION

A. BACKGROUND AND CONTEXT

THE presence of Unmanned Aerial Vehicles (UAVs), and particularly multirotor UAVs, has been remarkable in recent years in applications ranging from surveillance, aerial photography, to aerial manipulation, and transportation [1]–[4]. In particular, aerial manipulation has emerged as a prominent and rapidly evolving research area within the global control and robotics community [5]. Integrating manipulation mechanisms onboard unmanned aerial vehicles (UAVs), this paradigm enables in-flight grasping, transportation, and interaction with objects, significantly enhancing UAV functionality beyond conventional surveillance and monitoring tasks. Such physical interaction with the environment rep-

resents a qualitative shift in aerial robotics, demanding new levels of autonomy, precision, and robustness [6]. A first-level classification of the existing literature distinguishes between single-UAV and multi-UAV aerial manipulation systems [7]. In single-UAV architectures, the aerial platform is equipped with a manipulator or auxiliary mechanism to interact with the environment. Although these systems benefit from simpler hardware and control structures, they are inherently constrained by the limited payload capacity of an individual UAV, which restricts their applicability to lightweight and compact objects. In contrast, multi-UAV aerial manipulation relies on coordinated control of multiple aerial platforms to collectively manage a shared payload. In this context, each UAV is considered an agent; consequently, this subject falls

within the framework of cooperative control of multi-agent systems, which enables the execution of a wide range of coordinated tasks through local interaction among agents [8].

Another key criterion for categorizing aerial manipulation systems is the nature of the onboard manipulation mechanism. Initially, several strategies were considered for load transport, including a rigid connection between UAVs and the load using rigid robotic arms. The robotic arms allow object grasping and place operations, and they are mounted on the aerial vehicles that could be either actuated or underactuated [9]. However, having the load directly attached to the aerial vehicles significantly reduces their agility, and the coupling between translational and rotational dynamics makes end-effector operations more challenging [10].

Recently, new approaches have emerged in which a UAV performs transportation through a cable-suspended load [11]–[13]. The advantage of such a scheme is that it does not require additional components or actuators. Furthermore, it decouples rotational from translational dynamics, providing greater assurance of UAV integrity against potential load-induced dynamics during task execution [14]. Then, sophisticated control algorithms have been proposed for the transportation of individual cable-suspended loads. These techniques exploit differential flatness [15], passivity [16], and even differential geometry [17]. However, as mentioned, these approaches are inherently constrained by the limited payload capacity of individual UAVs.

Consequently, interest has arisen in the cooperative operation of multirotor UAVs to transport objects, using cables, that cannot be manipulated individually [18], [19]. While this paradigm significantly increases payload capacity and operational flexibility, it introduces substantial control and coordination challenges. To address these challenges, some groups have proposed evolved control techniques, such as centralized and decentralized linear Model Predictive Control (MPC) [20], Nonlinear Model Predictive Control (NMPC) [21], [22] or even Reinforcement Learning (RL) [23]. These approaches control the load's position and orientation, and integrate various safety constraints into the system's nonlinear dynamics [24]. While the NMPC has shown fascinating results, in all cases, the load's state (position, speed, orientation) is required in the control algorithm to meet the constraints imposed on the cost function. Furthermore, despite the significant advances in optimization methods, MPC, NMPC and RL remain computationally demanding for onboard control.

Less computationally demanding techniques have been explored. Some focus on tracking the trajectory of the load, considering both the feasibility of such a trajectory [25] and its asymptotic tracking [26], [27]. However, these works assume that the load's state (position and velocity) is known, which is a practical constraint. Other works focus on regulating the position of the UAV group and their associated formation geometry [28]–[30], and graph theory and multi-agent system control are partially leveraged [31]–[33].

Unfortunately, the techniques described above do not necessarily estimate and compensate in real time for the forces

and disturbances caused by the load and by interactions between UAVs. Motivated by these challenges, some approaches model the payload-induced effects as a lumped disturbance in the dynamics of the aerial robot formation, which must be estimated and compensated for. In the studies by [34], [35], a cooperative transportation based on master-slave paradigm of a rod by two UAVs is presented, implementing an admittance controller that operates with a force and torque estimator based on an Extended Kalman Filter (EKF) and a computationally demanding MPC for attitude and position of each UAV. A closely related methodology is presented in [36], where a decentralized adaptive control framework is employed to estimate an unknown payload mass by explicitly incorporating a force-consensus term. However, the approach is validated only in simulation. In [37], the cable tension at each UAV is measured using inertial sensors to estimate the state of the payload. The authors present a linear controller that works in conjunction with a two-stage Extended Kalman Filter (EKF) estimator, utilizing a leader-follower approach. In this setup, the leader UAV directs the system toward the goal location, while the follower UAV stabilizes the cable direction around the equilibrium point.

The approach proposed in the present paper is more in the spirit of [38], which presents a force-coordination control strategy to regulate both the formation and motion of two quadrotor UAVs transporting a cable-suspended pipe. The method introduces a force-consensus term derived from cable force estimation via a reduced-order Disturbance Observer (DO), and its effectiveness is experimentally validated. However, unlike our approach, which considers only the position to be shared with neighbors to reduce explicit communication, the consensus mechanism in [38] considers the position and velocity of the UAVs to ensure an equal distribution of payload mass while maintaining the formation within the horizontal plane. Furthermore, while [37] and [38] have a setup for two UAVs, the proposed approach in this paper can be generalized to N UAVs.

B. CONTRIBUTION

This work involves designing a cooperative load transportation system using quadrotors. The goal is to operate in unstructured environments, and due to various regulations in different countries, a human operator presence is required. However, because the system has a high number of degrees of freedom, the operator cannot simultaneously and accurately control all of them. Therefore, cooperative multi-agent control theory is proposed to design a control strategy that assists the operator. This control strategy enables the operator to focus solely on assigning, at the beginning of the operation and depending on the number of vehicles, the formation geometry and, later, providing the reference position to a single quadrotor within the group. Meanwhile, the formation geometry and the positions of the remaining agents are maintained automatically.

The present paper proposes to exploit the cascade architecture of the mathematical model of the quadrotor, hereafter

referred to as the agent. Using a disturbance observer and an appropriate thrust direction selection - via a rotation matrix and later mapped to the Euler angles - the perturbations induced by load transport are counteracted. Choosing these angles allows transforming the nonlinear dynamics into a system equivalent to three double integrators, corresponding to each independent axis. Thus, each agent incorporates a PI-structured velocity control with pre-compensation for acceleration, reducing the dynamics of each agent to three first-order integrators.

A distributed control is proposed under the well-known leader-follower consensus structure, allowing coordination, group formation, and collaborative load transport. In this scheme, the operator (pilot) acts as the leader agent, providing a reference position to a single member of the team.

Stability analysis of each control subsystem is presented, and the closed-loop multi-agent system stability is established in the Input-State Stability (ISS) sense, showing that the formation consensus error vector converges exponentially to a sphere centered at the origin of its state space, with a radius that can be arbitrarily reduced through proper selection of observer/control parameters and the algebraic connectivity of the communication topology.

The communication topology among agents is modeled as a connected, undirected graph, where each agent shares its position only with neighbors.

Finally, to experimentally validate the proposed strategy, a simple yet complete experimental platform is developed using: 1) open and standardized communication protocols, 2) the Pixhawk flight platform, and 3) a MATLAB-ROS (Robot Operating System) ground station.

The experimental results show that the mean squared error (MSE) for the formation drastically decreases when cable tension and direction estimation are included, demonstrating the feasibility of the proposed control strategy.

Additionally, the proposed control strategy was compared to the classical Leader-following formation that utilizes a "High-Gain" approach. The reliability of the control strategy was evaluated based on its abilities to reject disturbances and track waypoints, while also considering uncertainties in the UAV's mass.

Performance was measured using four widely recognized indices: the Integral of the Squared Error (ISE), the Integral of the Absolute Error (IAE), the Integral of Time Squared Error (ITSE), and the Integral of Time-weighted Absolute Error (ITAE). These results demonstrate the effectiveness, reliability, and importance of the proposed strategy.

The remainder of the paper is organized as follows: Section II presents the mathematical preliminaries of multi-agent systems and the mathematical model of the set of quadrotors transporting a point-mass load; Section III addresses the cooperative control design using the three main components mentioned above; Sections IV and V describe the development and main blocks of the experimental platform, as well as their corresponding results. Finally, Section VI presents the conclusions.

II. PRELIMINARIES AND MATHEMATICAL MODELING

A. GRAPH THEORY

Interactions in a multi-agent system can be represented using graphs. Consider the graph $\mathcal{G} = \{\mathcal{V}, \mathcal{E}\}$, consisting of a set $\mathcal{V}$ of N vertices (nodes) and edges $\mathcal{E}$. If there is an edge $(i, j) \in \mathcal{V} \times \mathcal{V}$ between nodes i and j , then i and j are called adjacent. $\mathcal{G}$ is called undirected or bidirectional if $(i, j) \in \mathcal{E} \Leftrightarrow (j, i) \in \mathcal{E}$; otherwise, it is called a directed graph. The adjacency matrix A is defined as $a_{ij} = 1$ if i and j are adjacent, and $a_{ij} = 0$ otherwise.

A path from i to j is a sequence of distinct nodes starting at i and ending at j , where each consecutive pair of nodes is adjacent. If there exists a path from i to j , then i and j are called connected. If all node pairs in $\mathcal{G}$ are connected and $\mathcal{G}$ is directed, it is called strongly connected. For bidirectional graphs, the term *strongly* is often omitted. The degree matrix D of $\mathcal{G}$ is a diagonal matrix with elements d_i equal to the cardinality of the neighbor set of node i , $N_i = \{j \in \mathcal{V} : (i, j) \in \mathcal{E}\}$. The Laplacian matrix $\mathcal{L}$ is defined as $\mathcal{L} = D - A$.

In the context of multi-agent system control, the leader-follower consensus configuration is of practical interest. In this setup, a leader is represented by an extra node $i = 0$, and information is exchanged between the leader and its neighboring nodes. Such interaction is represented by the graph $\bar{\mathcal{G}}$, which includes node $i = 0$ and the edges connecting the leader to its neighbors. The interaction between the leader and its neighbors is also represented by the diagonal matrix $\mathbf{G} = \text{diag}(g_1, \dots, g_N)$, where $g_i = 1$ if the i -th node is connected to the leader and $g_i = 0$ otherwise.

Here, we use the following useful lemma [31]:

Lemma 1. *The matrix $\mathbf{H} = \mathcal{L} + \mathbf{G}$, also called pinned Laplacian, corresponding to the graph $\bar{\mathcal{G}}$ has the following properties:*

- $\mathbf{H}$ has positive eigenvalues
- $\mathbf{H}$ is positive definite

B. MULTI-AGENT SYSTEM MODEL WITH LOAD

This work considers a system composed of N agents and a point-mass load of mass m_L , as depicted in Fig. 1. The following reference frames are used: $\mathbf{E}^f$ denotes the system's inertial frame with unit vectors $\{\mathbf{e}_x^f, \mathbf{e}_y^f, \mathbf{e}_z^f\}$, and $\mathbf{E}^i$ is attached to the center of mass of the i -th agent, with unit vectors $\{\mathbf{e}_x^i, \mathbf{e}_y^i, \mathbf{e}_z^i\}$.

Thus, the i -th agent relative to $\mathbf{E}^f$ has the position vectors $\mathbf{p}_i = (x_i, y_i, z_i)^T$ whilst $\mathbf{v}_i$ is its respective velocity and the load position is given by $\mathbf{p}_L = (x_L, y_L, z_L)^T$. Each agent is connected to the load via a constant length massless cable L_0 . The unit vector $\boldsymbol{\rho}_i := \frac{\mathbf{p}_i - \mathbf{p}_L}{\|\mathbf{p}_i - \mathbf{p}_L\|}$ indicates the direction from the i -th agent to the load.

The translational and rotational dynamics of each agent are

given by [39], [40]:

$$\Sigma_{P_i} : \begin{cases} \dot{\mathbf{p}}_i = \mathbf{v}_i \\ m_i \dot{\mathbf{v}}_i = -m_i g \mathbf{e}_z^f + \mathbf{R}_i^f T_i \mathbf{e}_z^i - \rho_i f_i \end{cases} \quad (1)$$

$$\Sigma_{R_i} : \begin{cases} \dot{\mathbf{R}}_i^f = \mathbf{R}_i^f [\boldsymbol{\Omega}_i^\times] \\ \mathbf{J}_i \boldsymbol{\Omega}_i = -[\boldsymbol{\Omega}_i^\times] \mathbf{J}_i \boldsymbol{\Omega}_i + \boldsymbol{\Gamma}_i + \boldsymbol{\Gamma}_d \end{cases} \quad (2)$$

where T_i is the thrust generated by the rotors along $\mathbf{e}_{z,f}^i$, f_i is the magnitude (with direction ρ_i) of the cable tension connecting the i -th agent of mass m_i to the load. The gravity vector is $g\mathbf{e}_z^f$, and $\mathbf{R}_i^f \in SO(3)$ is the rotation matrix projecting vector from $\mathbf{E}^i$ to $\mathbf{E}^f$.

The angular velocity of the agent i -th is $\boldsymbol{\Omega}_i$, with the inertia matrix $\mathbf{J}_i$, and $[\boldsymbol{\Omega}_i^\times]$ is its skew-symmetric matrix. $\boldsymbol{\Gamma}_i$ is the torque vector generated by the rotors and $\boldsymbol{\Gamma}_d$ is a perturbation torque vector. The system described by equations (1) and (2) is naturally cascaded in the sense that the rotational dynamic, that depends on the rotation matrix, is independent of translation. In our specific case, each subsystem has different response times ($\Sigma_{P_i} \approx 1s$, $\Sigma_{R_i} \approx 0.2s$), creating a natural time-scale separation [41], and therefore each subsystem is controlled separately.

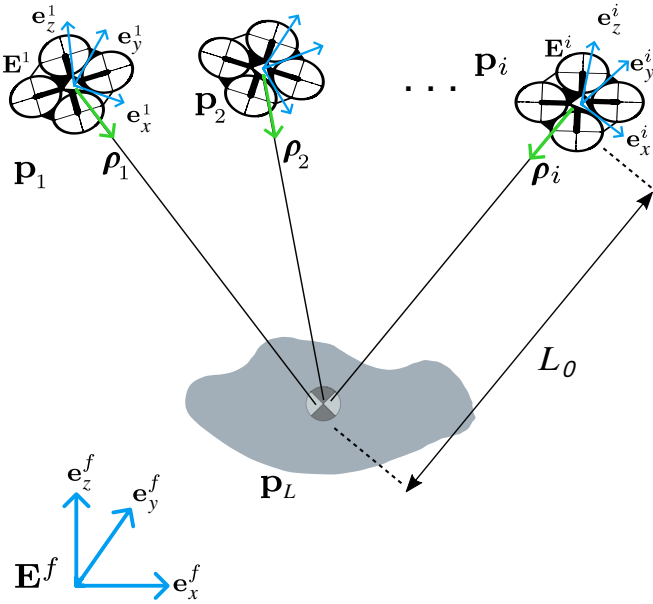


FIGURE 1. Representation of the i -th agent connected to a point mass m_L via a cable of length L_0 .

The load, modeled as a point mass, satisfies the dynamics [40]:

$$\Sigma_{P_L} : \begin{cases} \dot{\mathbf{p}}_L = \mathbf{v}_L \\ m_L \dot{\mathbf{v}}_L = -m_L g \mathbf{e}_z^f + \sum_{i=1}^n \rho_i f_i \end{cases} \quad (3)$$

The tension in the i -th cable can be modeled by Hooke's law with restitution constant $k_c \in \mathbb{R}_{>0}$ as follows:

$$f_i = \begin{cases} k_c (\underbrace{\|\mathbf{p}_i - \mathbf{p}_L\|}_{\Delta_{iL}} - L_0), & \text{if } (\Delta_{iL} - L_0) > 0 \\ 0, & \text{otherwise} \end{cases}$$

with direction given by ρ_i .

III. CONTROL STRATEGY

In this section, a collaborative control strategy for point-mass load transportation is presented, whose structure is illustrated in Fig. 2.

The control design exploits the cascaded structure given by equations (1) and (2), under the following assumption.

Assumption 1. Consider the rotational dynamics (2). It is assumed that the onboard autopilot¹ of each agent includes an attitude control algorithm capable of generating the control signal $\boldsymbol{\Gamma}_i$ such that $\mathbf{R}_i^f \rightarrow {}^d\mathbf{R}_i^f$ as $t \rightarrow \infty$, where ${}^d\mathbf{R}_i^f \in SO(3)$ is the desired orientation.

The first step is to stabilize the orientation of the i -th agent to a desired orientation ${}^d\mathbf{R}_i^f$ which must satisfy the following constraint:

$${}^d\mathbf{R}_i^f \mathbf{e}_z^i = \frac{m_i \mathbf{u}_i + m_i g \mathbf{e}_z^f + m_i \hat{\mathbf{z}}_{i2}}{\|m_i \mathbf{u}_i + m_i g \mathbf{e}_z^f + m_i \hat{\mathbf{z}}_{i2}\|} \quad (4)$$

where $\mathbf{u}_i$ is a control vector to be designed later to stabilize the linear velocity of each agent. The vector $\hat{\mathbf{z}}_{i2}$ represents the estimated tension and direction of the cable connecting the i -th agent of mass m_i with the load of mass m_L . This estimation is performed using the observer described in the next part.

A. CABLE TENSION AND DIRECTION OBSERVER

For the observer design, the following assumptions are made:

- The velocity $\mathbf{v}_i$ and orientation $\mathbf{R}_i^f$ are measurable²
- The nominal mass m_i of the agent is known;
- $\xi_i = \rho_i f_i / m_i$ with derivative $\dot{\xi}_i$ uniformly bounded, i.e., $\sup_t \|\dot{\xi}_i\| \leq K_0$, with $K_0 \in \mathbb{R}_+$

When the state vector is written as $\mathbf{z}_i := (\mathbf{z}_{i1}^T \mathbf{z}_{i2}^T)^T = (\mathbf{v}_i^T \xi_i^T)^T \in \mathbb{R}^6$, the translational velocity dynamics (equation (1)) and the perturbation dynamics can be written as:

$$\begin{cases} \dot{\mathbf{z}}_{i1} = -g \mathbf{e}_z^f + \frac{1}{m_i} \mathbf{R}_i^f T_i \mathbf{e}_z^i - \mathbf{z}_{i2} \\ \dot{\mathbf{z}}_{i2} = \dot{\xi}_i \end{cases} \quad (5)$$

Accordingly, the observer is proposed to estimate the perturbation vector:

$$\begin{cases} \dot{\hat{\mathbf{z}}}_{i1} = -g \mathbf{e}_z^f + \frac{1}{m_i} \mathbf{R}_i^f T_i \mathbf{e}_z^i - \hat{\mathbf{z}}_{i2} + \mathbf{L}_1 (\mathbf{z}_{i1} - \hat{\mathbf{z}}_{i1}) \\ \dot{\hat{\mathbf{z}}}_{i2} = \mathbf{L}_2 (\mathbf{z}_{i1} - \hat{\mathbf{z}}_{i1}) \end{cases} \quad (6)$$

The gain matrices $\mathbf{L}_1 = \text{diag}(l_1, l_1, l_1)$, $\mathbf{L}_2 = \text{diag}(l_2, l_2, l_2)$ are chosen such that the polynomial $P(\lambda) = \mathbf{I}_3 \lambda^2 + \mathbf{L}_1 \lambda + \mathbf{L}_2$ is Hurwitz.

Remark 1. In this work, a piecewise linear internal model is employed to describe the dynamics of additive disturbances. The experimental results described in Section V show that this was a good choice, even though, in practice, such disturbances could be highly time-varying.

¹<https://px4.io/>

²In practice, these variables can be obtained from: 1) the on-board navigation system within the autopilot (e.g., the PX4-based autopilot used in this work), which, typically, uses an Extended Kalman Filter (EKF) to provide high-quality/accuracy estimates. 2) the motion capture system. In this work, for simplicity, the second option was used.

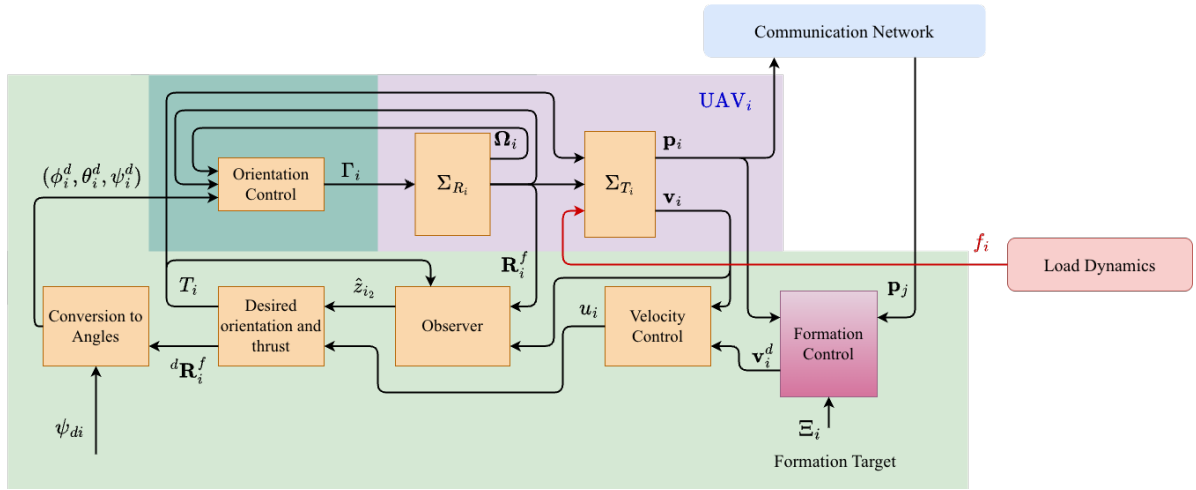


FIGURE 2. Block diagram illustrating the interaction between the i -th agent and the load. The external control loop (16) is represented as the Formation Control block. This block takes as input the signals $\mathbf{p}_i, \mathbf{p}_j$, and the formation specification for the i -th agent, denoted as Ξ_i . The output of this block is the desired velocity vector $\mathbf{v}_i^d$, which is then fed into the "Velocity Control" block where control law (10) is implemented. Additionally, there is an "Observer" block that resolves the dynamic system described by equation (6). This block receives the current orientation, linear velocity, and thrust of the i -th agent, specifically $\mathbf{R}_i^f, \mathbf{v}_i$, and T_i , respectively. The thrust T_i and the desired orientation—expressed as the rotation matrix ${}^d\mathbf{R}_i^f$ while adhering to constraint (4)—are calculated in the "Desired Orientation and Thrust" block. For ease of use, the desired orientation, constrained by (4), is converted into Tait-Bryan angles (a specific class of Euler angles) in the "Conversion to Angles" block. Finally, the calculated desired angles and thrust are transmitted to the onboard autopilot. Note that a communication network is assumed, enabling each agent to share its position and receive the positions of its neighbors for collaborative control.

Remark 2. For aggressive maneuvers, contact with obstacles or effects of cable parameters (k_c, L_0 , cable mass), disturbances could be considered highly time-varying; their internal model can be approximated, at least locally, by a time polynomial of arbitrary degree (e.g., parabolic, cubic, etc.). Consequently, a finite number of integrations extending the observer's state space could suffice to have an arbitrarily close estimation of the actual disturbance [42].

Proposition 1. Consider the dynamics in (5) and the observer in (6). Then, the estimation error $\tilde{\mathbf{z}}_i := \mathbf{z}_i - \hat{\mathbf{z}}_i$ satisfies the Input-to-State Stability (ISS) property [43], i.e., the solutions of the error dynamics converge to a sphere centered at the origin of the error state space with radius

$$\bar{\rho} = \frac{K_0}{\lambda_M} \quad (7)$$

where $\lambda_M \in \mathbb{R}_+$ depends on the chosen gain matrices $\mathbf{L}_1, \mathbf{L}_2$.

The proof is provided in the Appendix.

Thanks to attitude control (see Assumption 1), after a sufficiently long time, the agent's orientation converges to the desired orientation, $\mathbf{R}_i^f = {}^d\mathbf{R}_i^f$. Moreover, if the thrust T_i is selected as $T_i = \|m_i \mathbf{u}_i + m_i g \mathbf{e}_z^f + m_i \hat{\mathbf{z}}_{i2}\|$, the agent's orientation satisfies

$$\mathbf{R}_i^f \mathbf{e}_z^f = \frac{m_i (\mathbf{u}_i + g \mathbf{e}_z^f + \hat{\mathbf{z}}_{i2})}{T_i} \quad (8)$$

Substituting (8) into (1) gives

$$\dot{\mathbf{p}}_i = \mathbf{v}_i, \quad \dot{\mathbf{v}}_i = \mathbf{u}_i + \mathbf{y}_{zi} \quad (9)$$

with $\mathbf{y}_{zi} = \hat{\mathbf{z}}_{i2} - \mathbf{z}_{i2} = -\tilde{\mathbf{z}}_{i2}$.

B. COLLABORATIVE CONTROL FOR LOAD TRANSPORTATION

The control developed here follows an inner-loop/outer-loop structure. The inner loop regulates the velocity of the i -th agent, while the outer loop manages the position and coordination with neighboring agents.

Consider the second equation of (9) with the following control law:

$$\mathbf{u}_i = \dot{\mathbf{v}}_i^d + \mathbf{K}_D \tilde{\mathbf{v}}_{i2} + \mathbf{K}_I \tilde{\mathbf{v}}_{i1} \quad (10)$$

where

$$\tilde{\mathbf{v}}_{i2} = \mathbf{v}_i^d - \mathbf{v}_i, \quad \tilde{\mathbf{v}}_{i1} = \int_0^t \tilde{\mathbf{v}}_{i2} d\tau \quad (11)$$

and $\mathbf{v}_i^d$ is the desired velocity (to be designed later). The gain matrices $\mathbf{K}_D = \text{diag}(k_{d_x}, k_{d_y}, k_{d_z})$ and $\mathbf{K}_I = \text{diag}(k_{I_x}, k_{I_y}, k_{I_z})$ are chosen such that the polynomial $P_c(\lambda) = \mathbf{I}_3 \lambda^2 + \mathbf{K}_D \lambda + \mathbf{K}_I$ is Hurwitz.

The closed-loop error dynamics is

$$\underbrace{\begin{pmatrix} \dot{\tilde{\mathbf{v}}}_{i1} \\ \dot{\tilde{\mathbf{v}}}_{i2} \end{pmatrix}}_{\tilde{\mathbf{v}}_i} = \underbrace{\begin{pmatrix} \mathbf{0}_3 & \mathbf{I}_3 \\ -\mathbf{K}_I & -\mathbf{K}_D \end{pmatrix}}_{\tilde{\mathbf{A}}_{ci}} \underbrace{\begin{pmatrix} \tilde{\mathbf{v}}_{i1} \\ \tilde{\mathbf{v}}_{i2} \end{pmatrix}}_{\tilde{\mathbf{v}}_i} + \underbrace{\begin{pmatrix} \mathbf{0}_3 \\ -\mathbf{I}_3 \end{pmatrix}}_{\tilde{\mathbf{B}}_{ci}} \mathbf{y}_{zi} \quad (12)$$

Since $\mathbf{y}_{zi} = -\tilde{\mathbf{z}}_{i2}$ is uniformly bounded (see Proposition 1), $\sup_t \|\mathbf{y}_{zi}\|_\infty \leq \frac{K_0}{\lambda_M}$, we have the following result:

Proposition 2. Consider the dynamics in (12). The velocity error $\tilde{\mathbf{v}}_i := [\tilde{\mathbf{v}}_{i1}, \tilde{\mathbf{v}}_{i2}]^T$ satisfies the ISS property [43], i.e., the solutions converge to a sphere centered at the origin with radius

$$\bar{\rho}_z = \frac{K_0}{\lambda_M \lambda_{M_z}} \quad (13)$$

where $\lambda_{M_c} \in \mathbb{R}_+$ depends on the chosen gain matrices $\mathbf{K}_D, \mathbf{K}_I$.

The proof is provided in the Appendix.

After a sufficiently long time, the first equation of (9) becomes

$$\dot{\mathbf{p}}_i = \mathbf{v}_i^d + \mathbf{y}_{v_i} \quad (14)$$

where $\mathbf{y}_{v_i} = -\tilde{\mathbf{v}}_{i2}$ is the velocity error vector which, according to Proposition 2 is uniformly bounded, i.e., $\sup_t \|\mathbf{y}_{v_i}\|_\infty \leq \frac{K_0}{\lambda_M \lambda_{M_c}}$.

The last part of this section is to design the coordination control strategy, generating the desired velocity $\mathbf{v}_i^d$ of the i -th agent based on the positions of neighboring agents and a leader. The system follows a leader-follower configuration, where the leader is a human operator providing reference positions $\mathbf{p}_0(t_e) \in \mathbb{R}^3$ at times $t_e \in \{t_{e1}, t_{e2}, \dots, t_{em}\}$ to a limited subset of agents, enabling distributed control.

Proposition 3. Consider N agents with positions given by (14), interacting via a connected undirected graph $\mathcal{G}$. If the i -th agent communicates with the leader with constant $g_i > 0$, and the augmented graph including the leader is denoted $\tilde{\mathcal{G}}$, then for $K_P \in \mathbb{R}_{>0}$ the distributed control

$$\mathbf{v}_i^d = K_P \left[\sum_{j \in \mathcal{N}_i} (\mathbf{p}_j - \mathbf{p}_i) + g_i (\mathbf{p}_0 - \mathbf{p}_i) \right] \quad (15)$$

achieves leader-follower consensus in the ISS sense, i.e.,

$$\lim_{t \rightarrow \infty} \|\mathbf{p}_i(t) - \mathbf{p}_0(t)\| = \bar{\rho}_v, \quad \forall i \in \{1, \dots, N\}$$

for any initial condition $\mathbf{p}_i(t=0)$, where $\mathbf{p}_0 \in \mathbb{R}^3$ is the leader's state vector and $\bar{\rho}_v = \frac{K_0}{\lambda_M \lambda_{M_c} \lambda_{MLH}}$, with $\lambda_{MLH} \in \mathbb{R}_+$ depending on the algebraic connectivity of $\tilde{\mathcal{G}}$.

The proof is in the Appendix.

Remark 3. The convergence speed of the leader-follower consensus protocol is determined by the smallest eigenvalue of the pinned Laplacian matrix H , i.e., the largest eigenvalue of $-H$. A larger absolute value of this eigenvalue implies stronger network connectivity and faster propagation of the leader's information throughout the multi-agent system [31]. Furthermore, when disturbances are considered, the ISS stability analysis of the position error with respect to input disturbances provides the following insight: a larger eigenvalue implies faster convergence to zero and better disturbance rejection, as reflected in the ratio $1/\lambda_{MLH}$.

Practically, formation control is used instead of convergence to a single point to avoid collisions. Therefore, control law (15) is modified to

$$\mathbf{v}_i^d = \sum_{j \in \mathcal{N}_i} K_p (\mathbf{p}_j - \mathbf{p}_i) - K_p (\mathbf{p}_j^d - \mathbf{p}_i^d) + K_p g_i (\mathbf{p}_0 + \mathbf{p}_i^d - \mathbf{p}_i) \quad (16)$$

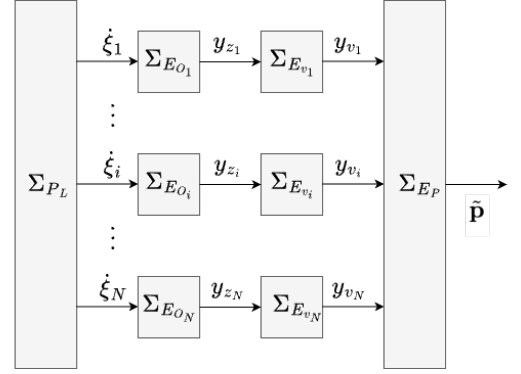


FIGURE 3. Representation of the closed-loop multi-agent system error dynamics.

where $\Xi = \{\mathbf{p}_1^d, \mathbf{p}_2^d, \dots, \mathbf{p}_N^d\}$ specifies the desired formation relative to the virtual leader $\mathbf{p}_0$. With this law, each agent i converges to its desired position $\mathbf{p}_0 + \mathbf{p}_i^d$, and the desired inter-agent distances are $\|\mathbf{p}_i^d - \mathbf{p}_j^d\|$. Defining the position error as $\tilde{\mathbf{p}}_i = \mathbf{p}_i - \mathbf{p}_0 - \mathbf{p}_i^d$, control (16) can be rewritten in the form of (21) in the appendix.

The convergence proof follows similarly to Proposition 3 and is omitted for brevity.

Fig. 2 shows the complete closed-loop control strategy.

Up to this point, the ISS stability property has been established for the observer, the velocity control, and the position control. The stability of the overall closed-loop system is stated in the following theorem.

Theorem 1. Consider the estimation, velocity, and position error dynamics, given by (19), (12), and (22), respectively.

$$\begin{aligned} \Sigma_{E_{O_i}} : & \begin{cases} \dot{\tilde{\mathbf{z}}}_i = \tilde{\mathbf{A}}_i \tilde{\mathbf{z}}_i + \tilde{\mathbf{B}}_i \dot{\xi}_i \\ \mathbf{y}_{z_i} = (\mathbf{0}_3 - \mathbf{I}_3) \tilde{\mathbf{z}}_i \end{cases} \\ \Sigma_{E_{v_i}} : & \begin{cases} \dot{\tilde{\mathbf{v}}}_i = \tilde{\mathbf{A}}_{c_i} \tilde{\mathbf{v}}_i + \tilde{\mathbf{B}}_{c_i} \mathbf{y}_{z_i} \\ \mathbf{y}_{v_i} = (\mathbf{0}_3 - \mathbf{I}_3) \tilde{\mathbf{v}}_i \end{cases} \\ \Sigma_{E_P} : & \begin{cases} \dot{\tilde{\mathbf{p}}} = -\mathbf{H} \tilde{\mathbf{p}} + \mathbf{y}_v \\ \mathbf{y}_p = \tilde{\mathbf{p}} \end{cases} \end{aligned}$$

Note that the error dynamic represents a cascade system (see Fig. 3). Therefore, the formation control of the multi-agent system for load transportation, composed of (6), (10), and (16), is ISS, when the i -th vector $\dot{\xi}_i$, $\forall i \in \{1, \dots, N\}$, is considered as input, and the formation error $\tilde{\mathbf{p}} = [\tilde{\mathbf{p}}_1^T \tilde{\mathbf{p}}_2^T \dots \tilde{\mathbf{p}}_N^T]^T \in \mathbb{R}^{3N}$ with $\tilde{\mathbf{p}}_i = \mathbf{p}_i - \mathbf{p}_0 - \mathbf{p}_i^d$ as state. Then $\tilde{\mathbf{p}}(t)$ converges to a bounded region determined by the input magnitude, i.e.,

$$\sup_{t \rightarrow \infty} \|\tilde{\mathbf{p}}(t)\| \leq \frac{K_0}{\lambda_M \lambda_{M_c} \lambda_{MLH}} \quad (17)$$

Proof. The proof is a direct consequence of the interconnection properties of systems that are ISS [43]. ■

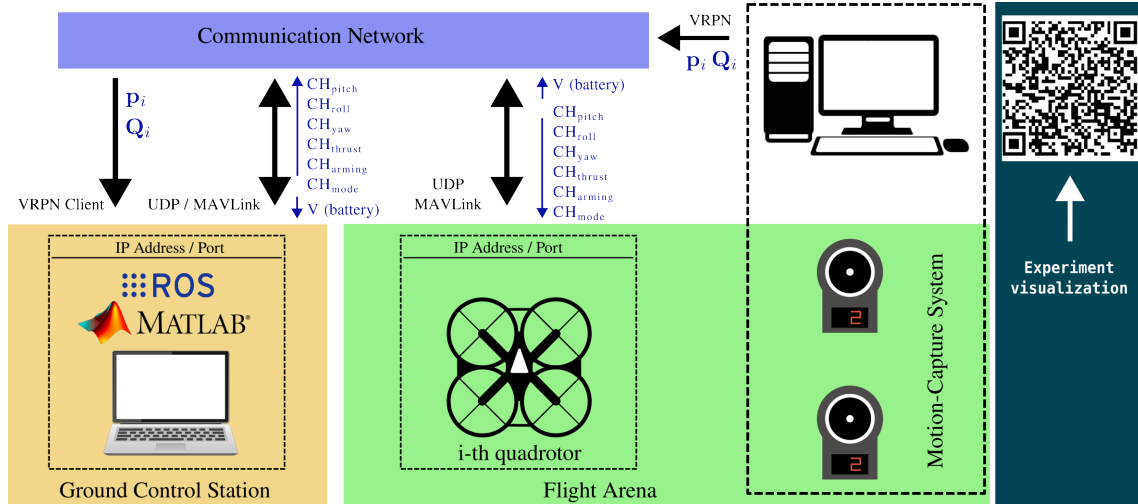


FIGURE 4. Modules and data flow of the experimental platform for drone control and monitoring. It consists of four workspaces: Wi-Fi communication network, ground control station, MoCap system, and flight test area. Key modules include: local router, ground station for executing control algorithms, MoCap equipment, and quadrotors. For a video of the experiment, scan the QR code.

IV. EXPERIMENTAL PLATFORM DESCRIPTION

To validate the control strategy, an ecosystem was developed for experimental deployment and analysis of results. This ecosystem consists of three main components: 1) a real-time motion capture (MoCap) system, 2) quadrotors equipped with an open-architecture autopilot, and 3) a ground control station responsible for i) processing information from the MoCap, ii) executing the control strategy, and iii) sending the desired orientation and thrust to each vehicle so that deployment, formation, and load transportation tasks are executed. Fig. 4 shows the described ecosystem.

A. MOTION CAPTURE SYSTEM

The localization stage is based on a motion capture system, comprising a network of infrared cameras and specialized software, which allows real-time monitoring of the movement of an object (vehicle) based on the light reflected by markers attached to the vehicle. In this case, the *Optitrack* system was used, which provides the *pose*, i.e., a position vector $\mathbf{p}_i \in \mathbb{R}^3$ and the attitude, which is parametrized by a unit quaternion $\mathbf{Q}_i \in \mathbb{S}^3$. To implement the disturbance observer (6), the velocity $\mathbf{v}_i$ is obtained by $\mathbf{p}_i$ and using a derivative filter. Furthermore, $\mathbf{R}_i^f$ is obtained by mapping $\mathbf{Q}_i$ to the parameterization of the corresponding rotation matrix [44].

B. OPEN-ARCHITECTURE UAVS

For research and development purposes, an open-source system was selected because it allows UAV programming and configuration, as well as communication using standardized open protocols. A PX4³-based autopilot was used, supporting Wi-Fi connection to the ground station via UDP (*User Datagram Protocol*), sending and receiving data packets through

MAVLink⁴ protocol. This enables operation similar to RC control, requiring at least four basic commands: *Roll* (ϕ_i^d), *Pitch* (θ_i^d), *Yaw* (ψ_i^d), and *Thrust* (T_i), corresponding to desired vehicle angles and thrust, while the internal attitude controller executes automatically. These desired angles and thrust are derived from the virtual control vector (4). The mapping between the desired rotation matrix and desired angles is omitted for space but can be found in [44].

C. GROUND CONTROL STATION

The ground station has the following features: a 7th-generation Core i5 dual-core 2.4 GHz processor, 16 GB RAM, Ubuntu 20.04 LTS, MATLAB 2023B. It serves as the central component of the ecosystem, managing communication through ROS nodes for data acquisition and UDP for command transmission, and running the observer (6) and control law (16). Upon its designation as the master node, the system receives and demultiplexes data from the Optitrack system, distributing it for processing by both the control system and the observer. The resultant control outputs are then converted into Pulse Width Modulation (PWM) signals to emulate traditional RC commands. Subsequently, these signals are packaged according to MAVLink protocols and transmitted over the network. For enhanced safety, the ground station incorporates additional commands that enable the remote activation and deactivation of the UAVs. Simultaneously, the PX4's dedicated software, installed on the Ground Control Station, displays real-time telemetry throughout the experimental operation.

⁴MAVLink (Micro Air Vehicle Link) is a protocol developed for communication with small UAV, providing access to control and telemetry, <https://github.com/mavlink/mavlink>.

³PX4 is an open-source autopilot system for low-cost UAV, <https://px4.io>.

D. EXPERIMENTAL PARAMETERS

Table 1 presents the values used in experiments. The gains for the outer-loop controller correspond to the cooperative control (16), while the inner-loop gains correspond to the individual velocity controller (10).

TABLE 1. Implementation Parameters

Variable	Value	Definition
m_i	0.526 kg	Vehicle mass
m_L	0.104 kg	Load mass
L_0	0.50 m	Cable length
K_P	0.95	Outer-loop control gain
(K_D, K_I)	(2.80, 0.41)	Inner-loop control gains
L	0.52	$\times 10^2$ Observer gain
	0	
	0	
	0	
	13.5	
	0	
	0	$\times 10^2$ Observer gain
	13.5	
	0	
	0	

V. EXPERIMENTAL RESULTS

A. EXP. 1: TAKE OFF AND LANDING OF TWO UAVS TRANSPORTING A LOAD

The first experiment involved the take-off and landing of the quadrotors with a suspended load of 104 g. In this scenario, the performance of the proposed control strategy is compared without and with estimated tension rejection, $\hat{z}_{i_2} = \hat{\xi}_i$, generated by the load in (4).

Once access from the ground station was established, the reference points p_0 of the virtual leader were generated in Cartesian coordinates. The specification of the desired formation of the two UAVs relative to the virtual leader p_0 is $p_1^d = [0; 0; 0]^T$ and $p_2^d = [0, 1, 0]^T$. The reference points for this experiment are:

$$\text{Exp.1} = \begin{cases} p_0(t = 0s) = [-0.5, 0, 0]^T, \\ p_0(t = 8s) = [-0.5, 0, 1]^T, \\ p_0(t = 70s) = [-0.5, 0, 0]^T. \end{cases}$$

With this configuration, the second agent has to maintain a 1 m separation along the y-axis relative to the first, while the load remains suspended approximately 50 cm between the two vehicles. Once take-off is executed, the tension along the y-axis produced by the suspended load causes the vehicles not to maintain the defined inter-distance. Additional self-disturbances due to air gusts induce oscillations around the reference points (see Fig. 5).

Focusing on the y-axis, where the load is shared, two opposing force vectors tend to pull the quadrotors together (see Fig. 6). It was observed that without compensating for the disturbance generated by the load, the vehicles collide in most trials due to the small gain K_p . By incorporating disturbance rejection based on estimation, real-time compensation of these perturbations is possible, significantly improving system stability. Quantitative results show that the mean squared error (MSE) between the current and desired inter-distance

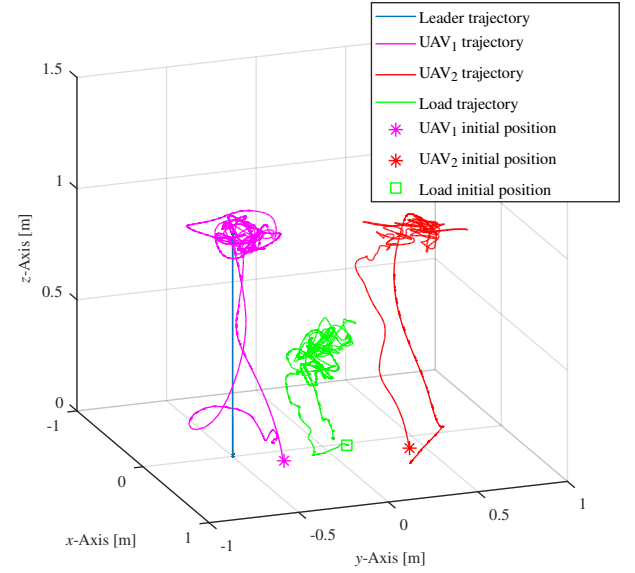


FIGURE 5. Take off and landing of agents transporting a 104 g payload.

between the UAVs (1m) decreases from 0.2664 m to 0.0324 m when disturbance rejection is included, demonstrating the effectiveness of the proposed control strategy. Experiment No. 1 is depicted in the first 48 s of the following video of the experiment.

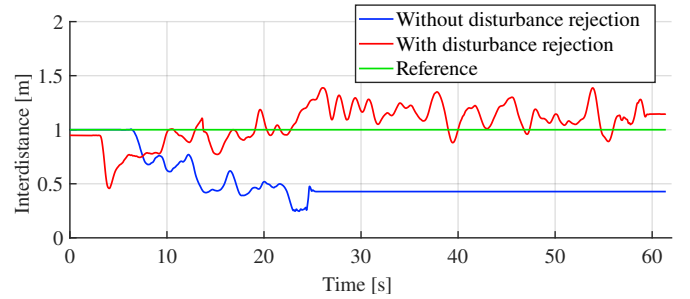


FIGURE 6. Comparison of the inter-distance evolution between the two quadrotors along the y-axis.

Fig. 7 shows the evolution of the estimated disturbance vector relative to the UAV 1. Due to the formation proposed in this experiment, the average estimated tension along the x-axis is zero, the mean force estimated along the z-axis corresponds to the load weight distributed between the two agents (≈ 0.510 N), and the y-axis force is the combination of the horizontal component of the weight and the force exerted by one quadrotor on the other.

Remark 4. Note that the collisions between the vehicles could be avoided without the use of the disturbance observer by increasing the proportional gain K_p in (16), yielding a simple controller for load transport. However, this elementary "High-Gain Solution" will produce aggressive movements, oscillations, or even instability, i.e., low reliability, as it will be show in V-C.

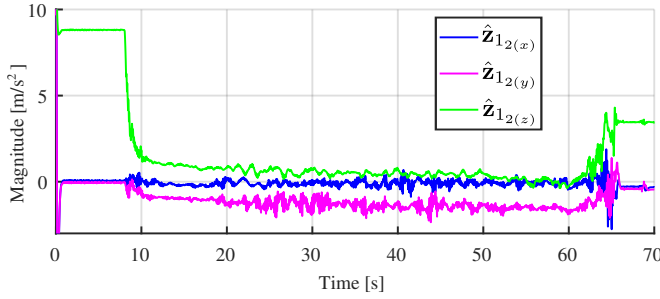


FIGURE 7. Acceleration estimated by the observer due to the cable tension on UAV₁ during the transport of a 104g suspended load.

B. EXP. 2: TRANSPORT OF A LOAD BY TWO UAVS

In the second experiment, the feasibility of transporting a load by two UAVs following a waypoint is evaluated. The reference points $\mathbf{p}_0$ correspond to the vertices of a 1 m square and are given by

$$\text{Exp.2} = \begin{cases} \mathbf{p}_0(t = 0\text{s}) = [0.5, -1, 0]^T, \\ \mathbf{p}_0(t = 8\text{s}) = [0.5, -1, 1]^T, \\ \mathbf{p}_0(t = 16\text{s}) = [0.5, 0, 1]^T, \\ \mathbf{p}_0(t = 24\text{s}) = [-0.5, 0, 1]^T, \\ \mathbf{p}_0(t = 32\text{s}) = [-0.5, -1, 1]^T, \\ \mathbf{p}_0(t = 40\text{s}) = [0.5, -1, 1]^T, \\ \mathbf{p}_0(t = 48\text{s}) = [0.5, -1, 0]^T. \end{cases}$$

As previously, the desired positions relative to the virtual leader $\mathbf{p}_0$ for the two agents are $\mathbf{p}_1^d = [0, 0, 0]^T$ and $\mathbf{p}_2^d = [0, 1, 0]^T$, respectively. With this configuration, the second agent maintains a 1 m separation along the y-axis relative to the first, while the load remains suspended approximately 50 cm between the two vehicles. This setup evaluates cooperative control and active disturbance rejection in a realistic scenario, where agents start from an initial point to complete the payload transport mission. Both inner and outer loops of the collaborative control are executed in MATLAB-Simulink. Since each vehicle has an assigned IP address, control signals and telemetry are monitored and stored in real time.

The leader's position follows a series of waypoints. These maneuvers involve accelerations that generate rapid and large-amplitude oscillations of the suspended load. The interest in this scenario lies in the increasing demand for agile autonomous aerial systems capable of operating effectively in critical and real-world environments [14], [25], as discussed at the beginning of this paper. In this context, the developed strategy cancels disturbances, demonstrating the feasibility and effectiveness of the technique (see Fig. 8).

Fig. 9 shows the evolution of the estimated forces on UAV₁ during the entire flight. The peaks correspond to changes in the leader's position. The estimated tension along z behaves similarly to the previous experiment, while in x and y, the estimated tensions increase due to oscillations.

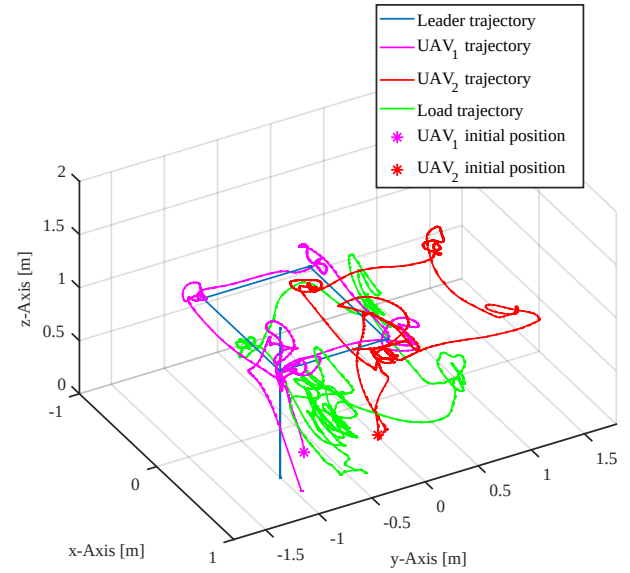


FIGURE 8. Load transportation mission with two quadrotors.

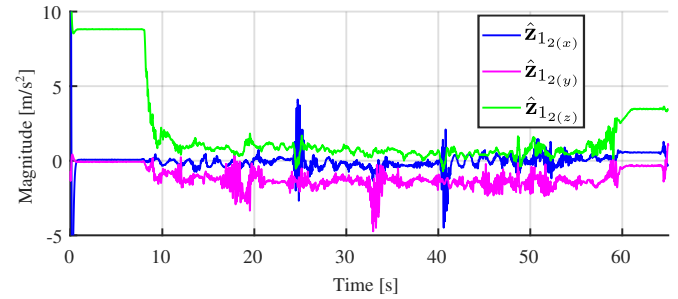


FIGURE 9. Acceleration estimated by the observer due to the cable tension on UAV₁ during the transport of a 104g suspended load.

C. EXP. 3: RELIABILITY EVALUATION FOR THE TRANSPORT OF A LOAD BY TWO UAVS

Reliability is evaluated through the disturbance rejection and waypoint tracking capabilities of the proposed control strategy, as well as the parameter uncertainties in the UAV mass. For that, the formation error is considered.

$$\tilde{\mathbf{p}}_i = \mathbf{p}_i - \mathbf{p}_0 - \mathbf{p}_i^d \quad \text{with } i \in \{1, 2\} \quad (18)$$

where $\mathbf{p}_1^d = [0, 0, 0]^T$ and $\mathbf{p}_2^d = [0, 1, 0]^T$ the waypoints $\mathbf{p}_0$ are similar to Exp. 2. The closed-loop with the proposed control strategy, as well as the classical Leader-following formation using "High-Gain" ($\mathbf{K}_p = 4.8$), was evaluated across 10 independent trials, each lasting 65 s and then averaged. Performance was quantified over the interval $t \in [10, 60]$ s using discrete-time forms of the four widely adopted indices: the Integral of the Squared Error (ISE), the Integral of the Absolute Error (IAE), the Integral Time Squared Error (ITSE), and the Integral of Time weighted Absolute Error (ITAE). The ISE penalizes large deviations more strongly due to the squared error term, highlighting the controller's ability to suppress significant tracking errors, while IAE evaluates the

accumulated magnitude of the tracking error and provides an overall measure of accuracy. Additionally, the time-weighted ITSE and ITAE indices emphasize errors that persist over time, thereby capturing the convergence and transient properties of the control strategy.

The results obtained show that the proposed control approach consistently achieves reliable waypoint tracking across all metrics (Table 2), whereas the simple leader-follower scheme using "High-Gain" exhibits larger accumulated errors and more persistent deviations as shown in Table 3. These results confirm that the proposed strategy improves both the accuracy and temporal convergence of the tracking performance.

	ISE	IAE	ITSE	ITAE
$\tilde{p}_{1x}$	0.8356	1.8733	26.561	56.606
$\tilde{p}_{1y}$	0.91093	2.2268	22.731	52.81
$\tilde{p}_{1z}$	0.8265	1.9573	22.672	53.097
$\tilde{p}_{2x}$	1.4242	2.7892	45.146	84.422
$\tilde{p}_{2y}$	1.4437	3.5465	34.215	79.363
$\tilde{p}_{2z}$	1.4611	2.9487	39.609	78.844

TABLE 2. Individual performance index considering the proposed control strategy.

	ISE	IAE	ITSE	ITAE
$\tilde{p}_{1x}$	1.1264	4.0882	33.239	115.35
$\tilde{p}_{1y}$	1.0532	4.2549	26.475	113.02
$\tilde{p}_{1z}$	1.9085	7.6557	49.743	211.72
$\tilde{p}_{2x}$	1.9384	5.824	56.489	164.7
$\tilde{p}_{2y}$	2.3912	8.166	59.553	222.01
$\tilde{p}_{2z}$	3.3781	10.593	87.194	292.13

TABLE 3. Individual performance index considering an elementary leader-following "High-Gain" solution.

Similarly to the aforementioned experiments, we evaluated the closed-loop reliability with respect to parameter uncertainties across 10 independent trials over a similar time span. Since the value of the UAV mass (m_i) is necessary for the observer implementation, we deliberately introduced an error of approximately 20% for the observer of UAV 1 and UAV 2. Table 4 represents the results of the four indices. It shows that this uncertainty produces a greater error along the z-axis, with the x- and y-axes unaffected. Then, the uncertainty in the UAV mass results in a position error in steady-state height. This was expected since the mass m_i is a parameter that affects only the third component of the first equation in (6).

	ISE	IAE	ITSE	ITAE
$\tilde{p}_{1x}$	0.8478	1.9759	26.854	60.459
$\tilde{p}_{1y}$	0.9022	2.3425	22.324	54.838
$\tilde{p}_{1z}$	0.9427	3.6966	26.752	105.62
$\tilde{p}_{2x}$	1.411	2.8463	44.625	86.155
$\tilde{p}_{2y}$	1.4481	3.8056	34.014	84.891
$\tilde{p}_{2z}$	3.1264	9.797	69.509	253.16

TABLE 4. Individual performance index considering uncertainties in the mass of the vehicles.

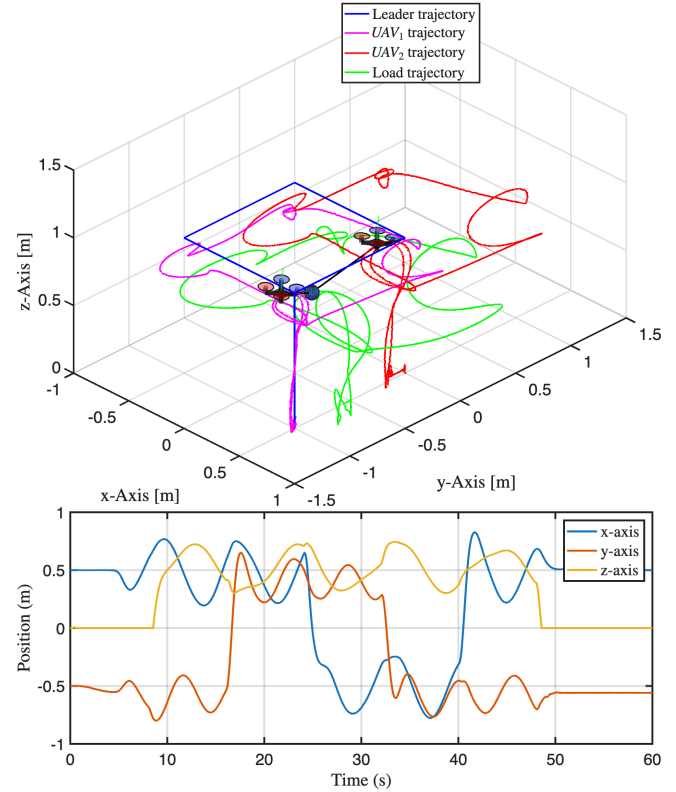


FIGURE 10. Payload behavior without observer

Figure (10) characterizes the payload dynamics in the absence of an observer. Figures (11) and (12) present the Euler angle trajectories of each UAV without observer implementation. Figure (13) depicts the thrust profiles generated by each UAV during the experimental trials. On the other hand, Figure (14) characterizes the system behavior under observer implementation. Figures (15) and (16) present the time evolution of the Euler angles for each UAV. Finally, Figure (17) depicts the thrust profiles generated by each UAV during the experimental trials.

In summary, the results corresponding to the observer implementation, as illustrated in the aforementioned figures, demonstrate an improved system performance compared to the case without an observer. The incorporation of the observer leads to enhanced tracking behavior and smoother system responses, as evidenced by the payload dynamics, Euler angle trajectories, and thrust profiles. These results highlight the effectiveness of the observer in improving the overall stability and performance of the UAV system transporting a payload.

VI. CONCLUSIONS

The work presented the design of a collaborative control strategy for load transportation using cable-suspended quadrotor UAVs. The control strategy was developed following an abstraction philosophy, decomposing the global problem into simpler subproblems and providing solutions for each. The

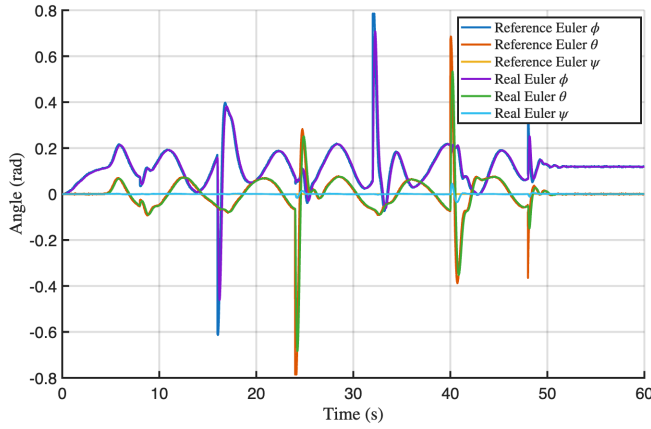


FIGURE 11. Euler angles to UAV 1 without observer

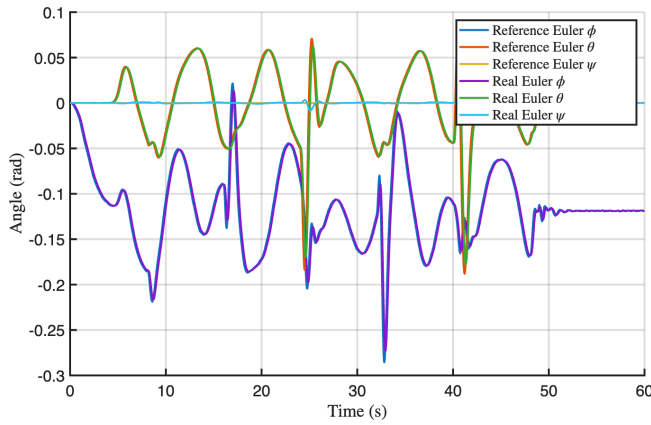


FIGURE 12. Euler angles to UAV 2 without observer

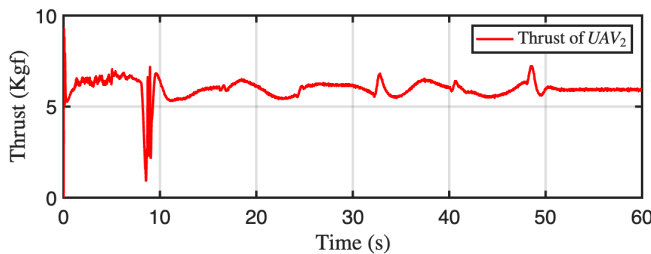
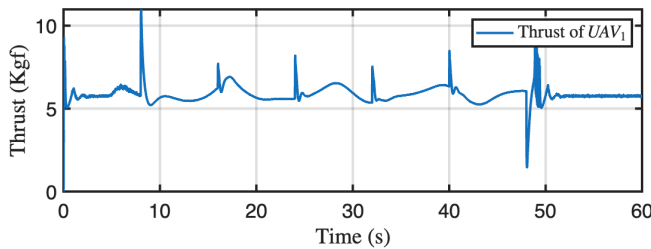


FIGURE 13. Thrust of UAVs without observer

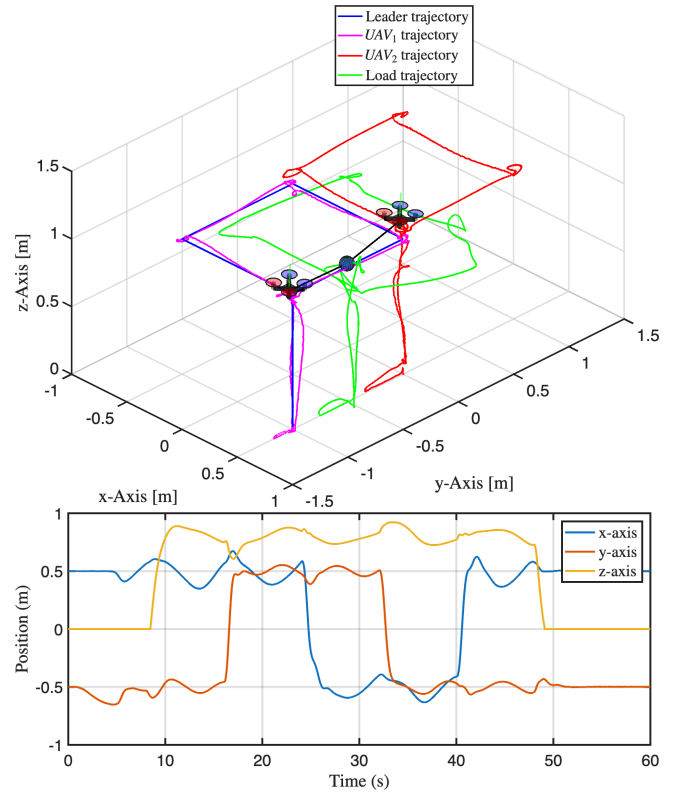


FIGURE 14. Payload behavior with observer

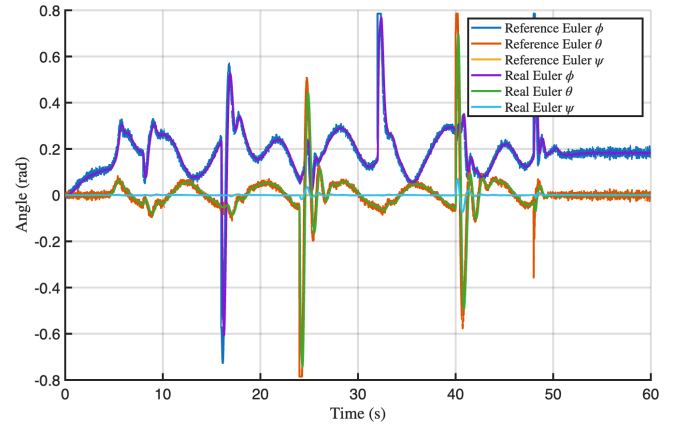


FIGURE 15. Euler angles to UAV 1 under observer working

core of the proposed approach lies in the real-time estimation and compensation of the cable tension forces transporting the load. Having demonstrated the effectiveness of the control strategy, future work will focus on testing the approach in outdoor environments and with a larger number of vehicles.

APPENDIX

A. PROOF OF PROPOSITION 1

Consider the estimation error $\tilde{\mathbf{z}}_i := \mathbf{z}_i - \hat{\mathbf{z}}_i$ and its derivative together with the dynamics described by (5) and the observer (6). Then, the estimation error satisfies the following differ-

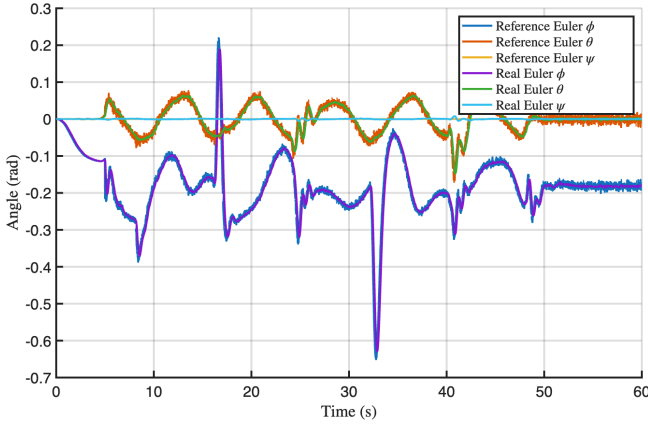


FIGURE 16. Euler angles to UAV 2 under observer working

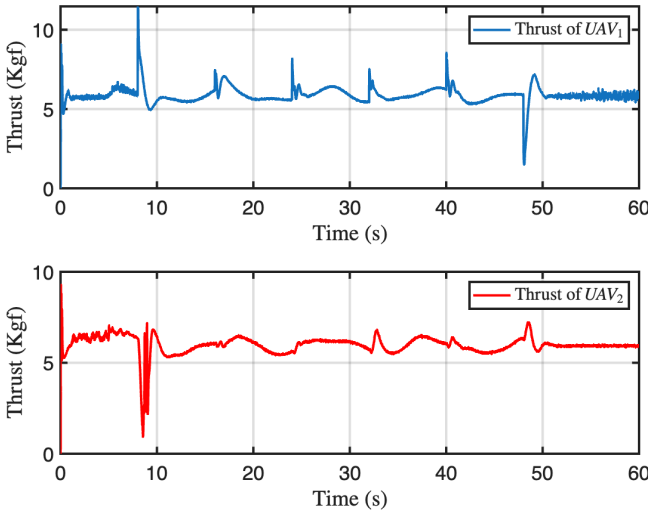


FIGURE 17. Thrust of UAVs under observer working

ential equation:

$$\begin{pmatrix} \dot{\tilde{z}}_{i1} \\ \dot{\tilde{z}}_{i2} \end{pmatrix} = \underbrace{\begin{pmatrix} -L_1 & I_3 \\ -L_2 & 0_3 \end{pmatrix}}_{\tilde{A}_i} \underbrace{\begin{pmatrix} \tilde{z}_{i1} \\ \tilde{z}_{i2} \end{pmatrix}}_{\tilde{z}_i} + \underbrace{\begin{pmatrix} 0_3 \\ I_3 \end{pmatrix}}_{\tilde{B}_i} \dot{\xi}_i \quad (19)$$

The characteristic polynomial of $\tilde{A}_i$ is $P(\lambda) = I_3\lambda^2 + L_1\lambda + L_2$. If the matrices L_1, L_2 are chosen such that $P(\lambda)$ is Hurwitz, then $\tilde{A}_i$ is Hurwitz, and the solution of (19) is bounded as follows:

$$\begin{aligned} \|\tilde{z}_i(t)\| &\leq \underbrace{\left\| \exp^{\tilde{A}_i t} \tilde{z}_i(0) \right\|}_{\beta(\|\tilde{z}(0)\|, t) \in \mathcal{KL}} + \underbrace{\|\xi_i\| \left\| \tilde{B}_i \right\| \int_0^t \left\| \exp^{\tilde{A}_i(t-\tau)} \right\| d\tau}_{\gamma(\|\xi\|_\infty) \in \mathcal{K}_\infty} \\ &\leq \exp^{-\lambda_M t} \|\tilde{z}_i(0)\| + \frac{K_0}{\lambda_M} (1 - \exp^{-\lambda_M t}) \end{aligned} \quad (20)$$

where $\lambda_M = \lambda_{\max}(\text{Sym}(\tilde{A}_i))$ is the largest eigenvalue of $\text{Sym}(\tilde{A}_i)$ with $\text{Sym}(\tilde{A}_i) = (1/2)(\tilde{A}_i + \tilde{A}_i^T)$ [45]. This demonstrates the existence of a class $\mathcal{KL}$ function β and a

class $\mathcal{K}$ function γ , indicating that the observer error dynamics is ISS, with the perturbation as input and the observation error as the state. This implies that the estimation error $\tilde{z}_i(t)$ converges exponentially to a sphere centered at the origin of its state space with radius $\bar{\rho} = \frac{K_0}{\lambda_M}$ as $t \rightarrow \infty$. Moreover, all trajectories starting within this sphere remain inside it. ■

B. PROOF OF PROPOSITION 2

The proof follows the same steps as Proposition 1. Therefore, the velocity error vector $\tilde{v}_i(t)$ converges exponentially to a sphere centered at the origin of its state space with radius $\bar{\rho}_z = \frac{K_0}{\lambda_M \lambda_{M_z}}$ as $t \rightarrow \infty$, where $\lambda_{M_z} = \lambda_{\max}(\text{Sym}(\tilde{A}_{c_i}))$. Additionally, all trajectories that begin within this sphere remain inside it. ■

C. PROOF OF PROPOSITION 3

Define the position errors of the i -th and j -th agents relative to the leader p_0 as $\tilde{p}_i = p_i - p_0$ and $\tilde{p}_j = p_j - p_0$, respectively. From (15), we have

$$\mathbf{v}_i^d = \mathbf{K}_P \left[\sum_{j \in \mathcal{N}_i} (\tilde{p}_j - \tilde{p}_i) - g_i \tilde{p}_i \right] \quad (21)$$

Using the global vectors $\tilde{\mathbf{p}} = [\tilde{p}_1^T \tilde{p}_2^T \dots \tilde{p}_N^T]^T \in \mathbb{R}^{3N}$, $\mathbf{y}_v = [\mathbf{y}_{v_1}^T \mathbf{y}_{v_2}^T \dots \mathbf{y}_{v_N}^T]^T \in \mathbb{R}^{3N}$, and $\mathbf{G} = \text{diag}(g_1, g_2, \dots, g_N) \in \mathbb{R}^{3N \times 3N}$, the closed-loop multi-agent position error dynamics are:

$$\dot{\tilde{\mathbf{p}}} = -\mathcal{L}\tilde{\mathbf{p}} - \mathbf{G}\tilde{\mathbf{p}} + \mathbf{y}_v = -\mathbf{H}\tilde{\mathbf{p}} + \mathbf{y}_v \quad (22)$$

From Lemma 1, the matrix $-\mathbf{H}$ is Hurwitz, and since the vector $\mathbf{y}_v$ is uniformly bounded by $\frac{K_0}{\lambda_M \lambda_{M_z}}$, the system (22) is ISS. That is, the consensus error vector $\tilde{\mathbf{p}}(t)$ converges exponentially to a sphere centered at the origin of its state space with radius $\bar{\rho}_v = \frac{K_0}{\lambda_M \lambda_{M_z} \lambda_{M_{LH}}}$ as $t \rightarrow \infty$. All trajectories starting within this sphere remain inside it. Here, $\lambda_{M_{LH}}$ is the largest eigenvalue of $\text{Sym}(-\mathbf{H})$, and its position in the complex plane depends on the algebraic connectivity of the graph $\tilde{\mathcal{G}}$ [31]. ■

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